

Collective Nudging that Scales.

How many dogs do I need to herd sheep?

1st Given Name Surname

name of organization (of Aff.), City, Country
email address or ORCID

2nd Given Name Surname

name of organization (of Aff.), City, Country
email address or ORCID

Abstract—Collective movement, such as flocking or swarming, facilitates nudging large groups via relatively few external entities, or shepherds – e.g. guiding drone swarms, controlling crowds or sheep-herding with robots. While some collective-control applications do explore scalability issues, a general study analysing scaling tendencies and the reasons behind them is still missing. This paper investigates: i) how the required number of shepherds scales with the number of individuals in the collective, given a particular flocking and shepherding scheme; and, ii) how such scaling tendencies correlate to the collective’s compactness, as an aggregate macro-measure. The study focuses on the case of sheep herding with dogs and takes an empirical approach via agent-based model simulations. Experimental results indicate: i) quasi-monotonically increasing, non-linear scaling of the number of dogs needed to herd growing flocks; coupled with diminishing returns and overcrowding effects when increasing the number of dogs for a given flock size; and, ii) strong correlation between the required number of dogs and the flock’s ‘spread’. These insights provide a basis for investigating the scaling tendencies of diverse collective-nudging applications, within different contexts.

Index Terms—scalability study, collective nudging, exogenous flock guidance, sheep herding, shepherd dogs, scaling break-point

I. INTRODUCTION

Collective movement [1], such as flocking or swarming [2], provides diverse benefits in both natural and artificial settings – e.g. safety in numbers for sheep flocks, cattle herds or fish schools [3],[4]; movement efficiency for bird flocks [5]; collective resilience and efficacy for ground robots or drone swarms [6]. The ensuing cohesion of such collective movement systems can also facilitate their control, or nudging, with relatively few external entities, or shepherds – e.g. herding of large sheep flocks with relatively few dogs [7], [8].

This insight opens valuable opportunities for natural, artificial or hybrid applications, where decentralised collectives may be guided to desired locations via exogenous controllers (i.e. external shepherds) rather than endogenous strategies (i.e. control algorithms internal to all individuals in the collective) – e.g. transporting ground robots to rescue- or cleaning-sites; guiding drone swarms through complex environments; assisting human groups in natural disaster situations.

The viability of decentralised strategies for collective nudging notably depends on their scalability with the number of individuals in the collective. Related proposals often consider scalability as a mere quality issue, for the particular application and context – e.g. [9] and [10] separately illustrate swarm con-

trol for ≈ 100 UAVs (via live demonstration and simulation); [7] show how a single dog can herd up to ≈ 100 sheep, with performance deteriorating afterwards. [11] emphasise the extra difficulty in nudging really small flocks, of up to 5 sheep.

To the best of our knowledge, a general study analysing scaling tendencies in collective nudging systems is still missing. This paper investigates how the number of shepherds increases with the number of individuals in the collective. It also correlates the observed shepherd scaling tendency with an aggregate macro-measure about the collective behaviour (i.e. ‘spread’) – as opposed to linking it to various combinations of individual behaviours and micro-configurations.

The study focuses on the case of sheep herding with shepherd dogs. This case was selected as sheep are highly gregarious animals, mimicking each other quite strongly when flocking [11]; and adopting sparser collective behaviours when grazing across well-known pastures [12]. This allows investigating the dogs’ efficacy in case of various sheep behaviours, leading to different flocking properties (e.g. compactness).

That said, we do not aim to provide a realistic model of sheep flocking or dog herding behaviour. Instead, for given flocking and shepherding algorithms (inspired by existing literature) we propose to estimate the resulting herding scalability – i.e. how the number of dogs (nudgers) increases with the number of sheep (flock size). While the concrete results will differ for different individual behaviours, the highlighted scaling principles and tendencies are likely to persist, within certain limits (i.e. as long as individual behaviours do not lead to radically different collective behaviours [2]).

Our study relies on agent-based model (ABM) simulations based on NetLogo platform¹. The tested herding task requires dogs to collect, drive and push sheep through a pen gate. Experiments determined task completion for different flock sizes (sheep numbers) and dog numbers. Results indicate a non-linear increase in the minimum number of dogs required to complete the task for growing sheep numbers. The scaling slope depends on the flock size range. Conversely, diminishing returns and overcrowding occur when adding more dogs for a given flock size. Finally, results reveal correlations between the number of dogs and the flock’s dispersal, or ‘spread’.

The main contributions of this paper include:

¹NetLogo agent-based platform: <http://ccl.northwestern.edu/netlogo/>

- 1) **Herding scalability study:** we analyse how the number of dogs must increase with the flock size (number of sheep) to successfully complete the herding task.
- 2) **Flocking macro-measure:** we correlate the scaling tendencies above with the flock’s distribution, or ‘spread’.
- 3) **Flocking model extension:** existing flocking algorithms assume that sheep can estimate their global centre of mass (GCM [3], [7]). We propose a model extension enabling sheep to approximate the flock’s GCM via a local indicator – i.e. sheep density gradient.

The presented scaling analysis suggests that the efficacy of each collective nudging setting (e.g. algorithms, configurations, contexts) is bounded within a limited scale interval – to be established case-by-case. Linking observed scalability tendencies to a collective macro-measure (e.g. mean-spread) allows to more easily transfer the specific results observed here to other collective nudging applications (e.g. featuring diverse behaviours, environments and goals).

II. RELATED WORK

A. Flocking models

Collective movement of gregarious animals has long been studied. [4] proposed the selfish herd theory according to which flock animals tend to pack tightly as a form of cover-seeking mechanism to lower individual predation risk by getting closer to the group’s centre. [1] showed how simple, local interaction rules can lead to group-level patterns; and how small parameter changes (e.g. orientation and attraction radius) can lead to sharp state-transitions. [13] found empirically that birds coordinate with a fixed number of topological neighbours ($n_{topo} = 7..8$) rather than with all neighbours within a metric distance. We rely on these principles to define our sheep flocking model, with $n_{topo} = 7$. Other swarming behaviours may rely on more or less neighbours and various local cues, e.g. $n_{topo} = 2..10$, depending on species [2]. Applications to artificial or hybrid systems have been studied within collective robotics areas [6], notably including drone swarms [9].

B. Shepherding

[7] introduced a ‘collect-or-drive’ heuristic, where a single dog alternates between collecting sheep outliers and driving the flock towards a goal. Sheep are mainly stationary with occasional random movements. Their flocking behaviour is based on attraction and repulsion forces. Authors note that sheep need to be attracted towards the centre of mass of at least half the flock for their algorithm to work. However, they do not model how individuals estimate this centre.

Similarly, [3] traced a sheep flock with GPS and noticed that they were attracted towards the flock centroid when under threat. They suggested that sheep might be unable to actually compute the flock centroid but could probably approximate it. This motivated our proposed model extension – enabling sheep to approximate their centre of mass by following a local density gradient, resulting from their attraction process.

[12] noted that sheep alternate between: slow grazing, when safe; and fast, “avalanche-like” regrouping when noticing a

predator. [14] used Ultra-Wideband tracking to study a border collie herding sheep. They confirmed that the dog would place itself behind the flock and adjust its heading depending on the flock direction. We adopted this trait in our model.

C. Scalable flock nudging

Most proposals of guided flocking or swarming systems discuss scaling issues as a quality attribute, tightly related to the targeted application. In the unmanned autonomous vehicles (UAV) literature, UAV groups are guided via endogenous algorithms (i.e. within the UAVs) and/or exogenous controllers (i.e. external to the collective). E.g., live trials during the DARPA OFFSET program demonstrated a single user directing over 100 UAVs (both physical and simulated) within an urban setting [9]. In a more theoretical setting, [10] proposed a self-organising agent-based control algorithm for drone traffic management. Simulations showed that collision risks increased supra-linearly with the number of drones, providing precise scaling figures for the particular settings tested.

For sheep herding, [7] studied how the efficacy of a single dog scaled with the flock size. They reported success rates of 100% up to 100 sheep, which then deteriorated to 80% for 200 sheep. To scale further, they also attempted splitting the flock into sub-flocks, yet with limited success. Moving to scenarios with several shepherds per flock, [15] showed that 2 to 5 shepherds were able to herd flocks of 10 to 40 members faster than a single one. They explored different herding strategies, such as line and arc formations for positioning shepherds behind the flock. We relied on their arc formation for the flock driving process in our model.

[16] studied structured, hierarchical flocks, with sheep alternating between ‘leader’ and ‘follower’ roles, resulting in intermittent grazing and collective movement behaviours. The study was conducted on tiny flocks (≤ 4 sheep); without shepherding. For really small flocks (≤ 5 members), [11] noted that sheep struggled to select a survival strategy, switching between solitary fleeing and selfish flocking behaviour [4]. This required higher shepherding skills, as featured in ‘sheepdog trial’² sporting competitions. We also observed performance degradation of dog herding for small flocks of 5-10 sheep.

[17] studied how shepherds could steer a group of evaders to a goal region. They derived scaling laws based on configurations of 2 to 25 shepherds and 4 to 625 targets. They found that the minimum number of shepherds N^* grows sub-linearly with the number of targets M (i.e., $N^* \propto \sqrt{M}$). They also noted that when target density falls below a minimum threshold M^{low} , the number of herders increases beyond $\sqrt{M}$. Their observation that sparser groups need more herders is consistent with our correlation between higher mean spread and lower success rates. However, their work does not assume any collective cohesiveness in the targets and consequently does not involve any flock-compacting phase.

[18] proposed a decentralised multi-robot herding strategy, where each robot targets the evader that is the furthest from the

²ISDS/HLF Sheepdog Archive: <https://www.isdssheepdogarchive.org.uk>

goal. They evaluated 1 to 5 robots guiding 30 to 100 evaders. The evaders would disperse in the absence of herders. They showed that, for more evaders, a smaller group of herders might be unable to accomplish the task. However, they did not derive scaling laws or correlate herding difficulty with a collective measure. Our work extends this to larger scales (up to 35 dogs and 400 sheep) and correlates the required number of dogs with a flock macro-measure (mean-spread).

III. GENERAL APPROACH AND CONTRIBUTIONS

A. Purpose and Research Questions

We aim to determine how the number of dogs required to complete the herding task varies with the number of sheep in the flock. To complete the herding task, dogs must: (p1) gather dispersed sheep into a compact flock; (p2) hold sheep within a containment zone for 800 time steps; (p3) drive sheep across an area and out through a pen gate. We are particularly interested in identifying scaling breakpoints – i.e., flock sizes past which a given number of dogs can no longer complete the herding task reliably. We analyse both scaling-up and -down tendencies, i.e. too few or too many dogs for a flock size.

This purpose leads to the following research questions:

RQ1: How does the viable interval of dog numbers $[D_{min}, D_{max}]$ required to accomplish the herding task vary with the number of sheep in the flock? This also implies:

RQ2: How can the required number of dogs be correlated with a flock-related measure (macro) rather than with sheep-related behaviours (micro)? To achieve realistic flocking, we also need: **RQ3:** How can sheep maintain global flock coherence (GCM attraction) based merely on local information?

B. Model Overview

1) *Herding flock model:* consists of N sheep herded by D dogs (Fig. 1). Sheep behaviour is inspired by the model in [7] (§ II-B); extended to use our density gradient for global flock cohesion. The dogs' collect-or-drive heuristic relies on [7], extended to support multiple dogs and patrolling during the holding phase (p2). Dog positioning behind the flock relies on [14] and the arc formation for multiple dogs on [15] (§ II-B).

2) *Flock mean-spread measure:* is the variance of sheep distances d_i to the flock GCM, averaged over time interval T :

$$\bar{S} = \frac{1}{T} \sum_{t=1}^T \text{Var}(\{d_i(t)\}_{i=1}^N) \quad (1)$$

The mean-spread $\bar{S}$ captures the dispersion of the flock around its centroid. Lower values indicate tighter, more compact flocks; whereas higher values suggest that many sheep are scattered from the centroid. While $\bar{S}$ does not measure flock density directly, it is correlated with it. A flock that is dispersed occupies more space and is sparser.

3) *Sheep-density gradient:* is our local indicator approximating the flock's centroid. As sheep only have local perception, they cannot directly compute the flock's actual GCM [3]. Instead, we propose that they move towards the densest neighbouring area. Namely, sheep scan their environment in 8

angular sectors (of 45°). They move towards the centre of the densest one (within their vision). An error of less than half a sector ($\leq |22.5|^\circ$) means that the sheep pick the correct sector.

C. Modelling Assumptions

Our model relies on the following assumptions:

- **Dog perception:** includes the overall flock position, allowing to estimate its GCM [7], [14]. Dogs can also focus on individual sheep, to detect outliers and whether they are handled by another dog.
- **Sheep perception:** is local and only includes $n_{topo} = 7$ neighbours for local cohesion [13].
- **Sheep vision:** is $\approx 360^\circ$. While real sheep have a $\approx 290^\circ$ to 313° vision field [19], we assume that head movements can compensate for the narrow rear blind zone.
- **Homogeneous sheep / dogs:** sharing the same behaviour (at any one time) – in reality they may differ [16].
- **Uniform world:** bounded, yet with no relief or obstacles.
- **No fatigue:** dogs and sheep have no energy limitations and no speed changes depending on physical effort.
- **Dogs' coordination:** self-partitioning the patrolling space around the flock – each dog taking a sector depending on its *id* and the total number of dogs.

D. Evaluation Patterns

To evaluate whether our research purpose is met (§ III-A), we first ensure that our flock herding model is 'realistic enough' relative to models from literature. For this, we ensure that our model reproduces some of the main behaviours discussed in § II. First, a single dog can reliably herd a flock of up to 100 sheep [7]. Second, in case of threat, sheep aim towards the flock's GCM, leading to a compact flock [3], [4]. Otherwise, they only cluster locally, leading to a sparser flock.

To evaluate our main purpose (RQ1), we take a qualitative approach. We assess the scaling tendency (e.g. supra-, sub- or linear) in the required number of dogs, depending on the flock size. To evaluate (RQ2), we assess whether the flock's 'spread' (i.e. compactness around the GCM) can predict the herding task difficulty. We expect that flocks that are more dispersed are harder to herd, requiring more dogs. To evaluate the auxiliary purpose (RQ3), we establish the difference between the sheep's ideal movement towards the GCM and the actual movement following our local density gradient.

IV. MODEL DESCRIPTION (ODD PROTOCOL)

We use the Overview, Design concepts and Details (ODD) protocol [20] to describe our model of a flock of N sheep herded by D dogs (Fig. 1). Time is simulated in discrete steps, or 'ticks'. Distances and radii are expressed in 'patches' and speeds in patches/tick. The herding task must be completed within a maximum of 10,000 time steps (ticks), including (p1) *sheep collecting*, (p2) *holding* and (p3) *exiting* (§ III-A).

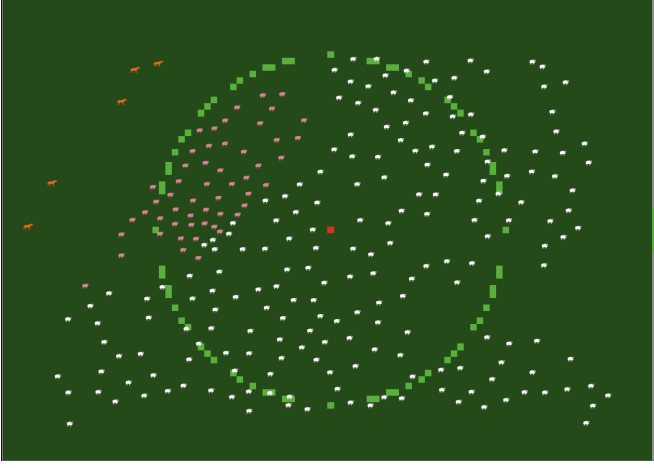


Fig. 1. Simulation environment illustration (for $D = 5$, $N = 250$). White sheep are grazing, pink ones are fleeing; dogs appear in orange (top-left). The green circle is the containment zone, with its centre at the red dot. The exit gate appears in green on the right grey wall.

A. Entities and State Variables

The main model entities are (Fig. 1):

- 1) **Dogs:** herding agents featuring a position, heading, mode $\in \{\text{targeting}, \text{driving}, \text{patrolling}\}$, target sheep, travelled distance, patrol angle and stuck counter.
- 2) **Sheep:** flocking agents featuring a position, heading, speed, mode $\in \{\text{grazing}, \text{fleeing}\}$, stuck counter.
- 3) **Environment:** a non-toroidal grid, including a circular *containment zone* at its centre and an *exit gate* in the middle of its right boundary.

B. Process Overview and Scheduling

The simulation proceeds at each tick, in order, as follows:

- 1) **Phase update:** the simulation goes through the three phases $\in \{\text{collecting}, \text{holding}, \text{exiting}\}$, sequentially, based on the flock's state (§ IV-D).
- 2) **Dog movement:** each dog evaluates the flock's state and reacts accordingly based on its own mode (§ IV-F).
- 3) **Sheep movement:** each sheep determines its mode $\in \{\text{grazing}, \text{fleeing}\}$ (§ IV-E2), computes its direction (§ IV-E3) and moves accordingly (§ IV-E9).
- 4) **Success evaluation:** when reaching the time limit (10,000 ticks), the simulation checks if the success conditions are met – i.e. all sheep exited through the gate (§ IV-D).
- 5) **Metric recording:** the flock's mean-spread is updated.

Within each breed (dogs, sheep), individual states are updated sequentially, in random order (to simulate parallelism).

C. Initialization

We simulate an environment grid of 101×71 patches; with an exit gate of 6 patches; and a containment zone of radius $r_c = \text{clamp}(2.5\sqrt{N}, 23, 27)$ – i.e. $r_c = 23$ for $N \leq 84$, $r_c = 2.5\sqrt{N}$ for $N \in [85 - 116]$ and $r_c = 27$ for $N \geq 117$. The exit gate only influences sheep behaviour, via an attraction force, during the exiting phase (p3). D dogs are placed in

a grid pattern in the top-left corner. N sheep are placed randomly within a 2-patch buffer from the environment's boundaries and a 5-patch buffer from any dog. Sheep start in grazing mode and dogs in targeting mode, all heading in random directions. Table A1 (Appendix A) lists all model parameters.

D. Phase-transition sub-model

The simulation goes through three subsequent phases: (p1) **Collecting:** dogs gather the scattered sheep and lead them towards the centre of the containment zone; (p2) **Holding:** dogs patrol the containment zone's perimeter to keep the flock within it; (p3) **Exiting:** dogs compact the flock and drive it towards the exit gate. The phase-transition conditions are:

• Collecting $\rightarrow$ Holding

- a) All the sheep must be within the containment zone;
- b) The flock centroid is within 15% of the containment zone's radius r_c relative to the zone's centre;
- c) There are no 'lost' sheep (i.e. farther than $\ell = 4\sqrt{N}$ from the flock centroid).

• Holding $\rightarrow$ Exiting

- a) The sheep have been withheld within the containment zone for 800 ticks;
- b) There have been no excessive escapes (i.e. $> 5\%$ sheep outside the zone for $D \geq 2$; and $> 10\%$ for $D = 1$). Otherwise, the simulation goes back to collecting.

• Exiting $\rightarrow$ Success: All sheep have passed through the exit gate within 10,000 ticks. If not, the herding task fails.

E. Sheep behaviour sub-model

1) *Modes:* determines sheep behaviour, as follows:

- *Grazing* sheep explore their environment at low speed ($v_{\text{graze}} = 0.01$), maintaining their heading through directional persistence, subject to some stochastic noise. They align with the average heading of their $n_{\text{topo}} = 7$ neighbours within their vision range (30 patches). Flock coherence is only maintained via a weak local attraction, with no global attraction towards the GCM. Inter-sheep repulsion is relatively large (compared to fleeing mode) and the resulting flock relatively loose, drifting slowly across the pasture. Table I lists the force weights and Table A1 the parameter values.
- *Fleeing* sheep move faster ($v_{\text{flee}} = 0.3$), the inter-sheep repulsion is reduced (stress multiplier σ , Table I) and the flock packs tighter. A global cohesion force attracts the sheep towards the densest visible sector, approximating the flock GCM. Directional persistence and alignment are suppressed and dog avoidance takes over.

2) *State transition:* triggers sheep into fleeing when:

- A dog is within a fleeing radius $R_{\text{flee}} = 14$, or
- A dog is within a vigilance radius $R_{\text{vig}} = (1.8 \times R_{\text{flee}}) = 25.2$ and at least 30% of the visible sheep n_{topo} are already fleeing. This is the fear contagion phenomena.

Sheep return to grazing mode when none of these conditions hold. Transitions are immediate.

3) *Force model*: dictates sheep movement. Each sheep computes a force vector $\mathbf{f}$, as a weighted sum of force components. Each component has a scalar magnitude F applied along a direction (e.g. away from a dog, towards neighbours). Forces are expressed as relative weights (without physical units). They define a desired heading. Components depend on the mode:

When grazing:

$$\mathbf{f}_{\text{grazing}} = W_{\text{rep}} * \sigma * \mathbf{f}_{\text{rep}} + W_{\text{local}} * \mathbf{f}_{\text{local}} + W_{\text{wall}} * \mathbf{f}_{\text{wall}} + W_{\text{persist}} * \mathbf{f}_{\text{persist}} + W_{\text{align}} * \mathbf{f}_{\text{align}} + \xi \quad (2)$$

When fleeing:

$$\mathbf{f}_{\text{fleeing}} = W_{\text{flee}} * \mathbf{f}_{\text{flee}} + W_{\text{rep}} * \sigma * \mathbf{f}_{\text{rep}} + W_{\text{local-flee}} * \mathbf{f}_{\text{local}} + W_{\text{global-flee}} * \mathbf{f}_{\text{global}} + W_{\text{wall}} * \mathbf{f}_{\text{wall}} + \mathbf{f}_{\text{gate}} + \xi \quad (3)$$

Where ξ is a stochastic noise term that is added to avoid deterministic behaviour [21], [7], [12].

Weight	Fleeing	Grazing
W_{flee} (dog avoidance)	3.5 (0.5 at gate)	–
W_{rep} (repulsion)	1.5	1.5
σ (stress multiplier)	0.5	1.0
W_{local} (local cohesion)	1.5	0.5
W_{global} (global cohesion)	0.75	–
W_{wall} (wall avoidance)	5.0	5.0
W_{persist} (persistence)	–	0.5
W_{align} (alignment)	–	0.3

TABLE I
FORCE WEIGHTS BY MODE.

4) *Dog avoidance*: repels sheep from dogs within a radius $R_{\text{flee}} = 14$. The force is inversely proportional to the distance from the dog: $F = (R_{\text{flee}} - d_{\text{sheep,dog}})/d_{\text{sheep,dog}}$. During exiting (p3), when a sheep is near the gate (within 3 horizontal and 5 vertical patches), the dog avoidance weight is reduced (Table I) to ease the sheep's exit and to avoid bouncing back.

5) *Sheep repulsion*: repels sheep from other sheep within a short range, $R_{\text{rep}} = 3$. The magnitude is $F = (R_{\text{rep}} - d_{\text{sheep1,sheep2}})/d_{\text{sheep1,sheep2}}$; reduced by σ when fleeing.

6) *Cohesion*: occurs simultaneously at two levels:

- *Local cohesion* attracts sheep towards the local centre of mass (LCM) – i.e., the centroid of the n_{topo} neighbours. When fleeing, force f_{local} activates when the distance to the LCM is greater than R_{rep} . When grazing, the activation threshold is $2 \times R_{\text{rep}}$ and the weight is weaker.
- *Global cohesion* attracts fleeing sheep towards the flock's GCM [3], relying on the density gradient (§ III-B3).

7) *Gate attraction*: is detailed in Appendix B.

8) *Wall avoidance*: is an inverse-square repulsion within a radius of 8.0 patches from the wall, with $F = 1/(d_{\text{wall}}^2 + \epsilon)$, $\epsilon = 0.01$. It is suppressed near the gate during exiting.

9) *Sheep movement*: limits a sheep's direction change to 40° per tick. Grazing follows a distribution similar to Lévy [22] with $v = \min(0.2, 0.01 \times e^U)$, $U \sim \text{Uniform}(0, 5)$. Hence, grazing sheep are mainly still, with occasional longer steps [7]. The fleeing speed v_{flee} is 0.3 patches per tick and the noise ξ is ± 0.2 per tick in each direction (Tables I& A1).

F. Dog behaviour sub-model

Dogs are subject to inter-dog repulsion, sheep avoidance, wall avoidance, speed modulation near other agents and getting-stuck detection (Appendix C). In addition, dog positioning depends on one of their three modes (Table II):

Mode	Active during	Trigger
Targeting	Collecting, Exiting	There are lost sheep
Driving	Collecting, Exiting	No lost sheep
Patrolling	Holding	Flock contained

TABLE II
DOG MODES AND HOW THEY ACTIVATE.

1) *Targeting*: allows each dog to select the nearest 'lost' sheep. The dog positions itself behind the target relative to the flock centroid (at $d_{\text{behind}} = 9$ patches). When within $d_{\text{curve}} = 25$ patches, the dog curves its approach to avoid scaring the target away from the flock and to nudge it towards it. If all outliers are assigned, the dog stays idle to not disturb the flock.

2) *Driving*: occurs when the flock is compact and there are no outliers. The dogs position in a 90° arc formation behind the flock centroid, in the opposite direction to the movement target, at the distance $d_{\text{push}} = 9$. The target direction depends on the simulation phase. During collecting, dogs aim for the centre of the containment zone; while during exiting, they target the gate. If more than 10% of the flock is 'lost' and the dog is within $1.1 \times d_{\text{push}}$ of the flock centroid, the dog retreats at half speed – to avoid further splitting the flock.

3) *Patrolling*: only occurs during the holding phase. The containment zone is divided into equal angular sectors. Each sector is matched to a single dog that will patrol its sector at 5 patches from the containment zone (or 8 when the containment zone radius $r_c < 25$). Each dog tracks the sheep that are closest to the boundary of the containment zone, within its sector. Dogs no longer enter the containment zone.

V. EXPERIMENTS

A. Experimental Design

We conducted experiments varying two factors:

- **Number of dogs**: $D \in \{1, 2, 3, 4, 6, 10, 15, 20, 25, 35\}$.
- **Number of sheep**: $N \in \{5, 10, 25, 50, 100, 150, 200, 250, 300, 350, 400\}$.

This gives $10 \times 11 = 110$ experimental conditions, that were run 100 times each, for a total of 11,000 simulations. The other parameters were constant (Table A1). Each run terminated when either: all sheep exited; or after the 10,000 ticks timeout.

B. Metrics

We recorded key metrics for each run:

- **Success**: boolean indicating herding-task completion;
- **Ticks to success**: time taken by successful herding tasks;
- **Current phase**: the phase at the end of the simulation;
- **Sheep in zone** and **Holding time**: how many sheep were contained and for how long, respectively;

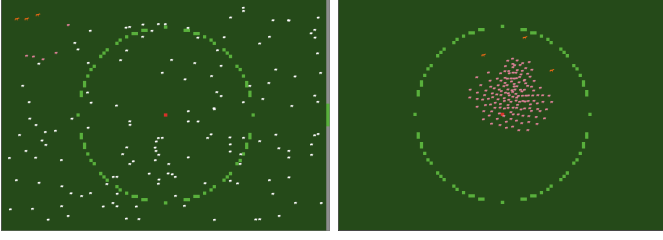


Fig. 2. Example of experimental condition ($D = 3$, $N = 150$) with different spreads: Left: dispersed flock ($\bar{S} = 164.4$). Right: compact flock ($\bar{S} = 9.3$).

- **Mean spread:** the average of sheep distances to the flock centroid, over the entire simulation (eq. 1). Compact flocks feature $\bar{S} \approx 15$ or lower; and sparse flocks $\bar{S} \approx 50$ or higher. Fig. 2 illustrates the contrast for a specific condition ($D = 3$, $N = 150$), with $\bar{S} = 164.4$ when the flock is dispersed and $\bar{S} = 9.3$ when it is compacted.
- **Total dog distance:** sum of distances covered by all dogs;
- **Lost sheep:** outliers at the end of the simulation;
- **Enclosed sheep:** Number of sheep that passed the gate.

C. Implementation

The model was implemented and tested using NetLogo7.0.3.

VI. RESULTS

A. Success Rates, Scaling Breakpoints and Viable Ranges

Table III presents the success rates (SR) for all (D, N) conditions. For a single dog ($D=1$), we observe a sharp transition from near total success (99%-100%) to near total failure (1%) when enlarging the flock from 100 to 150 sheep – i.e. ≈ 100 sheep is the scaling breakpoint for one dog. For more dogs, the transition between total success and mostly failure is more progressive, including several intermediate success rates (15%-90%). These transition zones become wider as the number of dogs increases (e.g. 1 flock size for $D = 2, 3, 4$ and 5 sizes for $D = 15$). For $D > 15$, the maximum flock size ($N = 400$) was insufficient to cause persistent failure.

Adding more dogs shifts the breakpoint to larger flocks (Figure 3, Appendix E). However, the return per additional dog diminishes. To ensure success ($SR > 90\%$), a dog can handle up to $N=100$ sheep (1 dog : 100 sheep); yet we need 3 dogs for 150 sheep (1:50); and 20 dogs for 200 sheep (1:10). Reaching $SR > 50\%$ success for $N=400$ requires 20 dogs (1:20), but reaching $SR > 90\%$ requires 35 dogs (1: ≈ 11). The growing gap between breakpoints suggests that larger flocks are increasingly difficult to herd. Conversely, for smaller flocks, employing too many dogs leads to *overcrowding* and lower SR (e.g. for $N=10$, $D \geq 20$ performs worse than $D < 20$).

This suggests that for a given flock, scaling breakpoints may occur both for insufficient and superfluous dog numbers. This gives a viable dog-scaling range $[D_{min}, D_{max}]$ (Table A3, Fig.4). The range varies for diverse flock sizes, probably due to non-linear interactions between different numbers of dogs and sheep. E.g., the widest range occurs for $N=100$, where all tested dog numbers $D=[1, 35]$ reach success. Conversely, for $N=200$, only two values $D \in \{20, 25\}$ reach $SR > 90\%$ success.

$N \backslash D$	1	2	3	4	6	10	15	20	25	35
5	100	100	100	98	100	100	99	99	98	99
10	99	100	100	100	99	99	97	86	85	70
25	100	100	100	99	99	98	98	98	93	88
50	100	100	100	99	99	99	99	97	94	81
100	100	100	100	100	100	100	98	95	92	95
150	1	80	99	97	96	95	97	97	97	92
200	0	9	57	68	71	89	87	93	94	89
250	0	0	5	11	46	71	85	89	93	91
300	0	0	0	1	10	50	74	93	93	90
350	0	0	0	0	3	34	64	85	86	91
400	0	0	0	0	0	6	46	61	85	91

TABLE III
SUCCESS RATE, SR (%), PER CONDITION (100 RUNS EACH)
WITH $\geq 90\%$, 15–90%, $< 15\%$

$N \backslash D$	1	2	3	4	6	10	15	20	25	35
5	42.0	20.6	14.1	13.1	12.2	13.4	11.6	10.4	13.0	12.2
10	28.1	15.9	15.7	14.3	15.2	12.7	14.2	16.9	18.0	21.4
25	17.6	16.1	15.7	14.5	14.3	15.2	14.5	14.7	15.8	18.2
50	21.2	18.5	17.3	15.9	15.4	15.6	15.5	16.7	17.0	17.1
100	27.3	21.8	16.6	15.7	14.5	13.9	14.3	17.1	15.7	16.9
150	49.3	23.4	18.8	20.6	15.0	15.6	15.6	16.2	16.0	17.7
200	63.5	39.1	23.1	18.7	15.6	15.8	16.2	16.6	16.9	18.9
250	70.0	57.1	37.4	31.7	17.4	16.5	17.3	17.6	17.5	19.0
300	76.0	71.6	60.2	43.0	28.5	17.6	17.9	18.7	18.2	19.4
350	83.6	79.4	76.6	65.7	36.6	19.2	18.4	18.7	19.6	20.1
400	92.1	87.0	84.0	80.0	56.5	22.0	19.5	19.3	20.3	20.8

TABLE IV
MEAN SPREAD $\bar{S}$ AVERAGED OVER ALL RUNS
WITH SUCCESS RATES $\geq 90\%$, 15–90%, $< 15\%$

B. Success Prediction via the Mean-Spread

We correlate the above success rates (SR) with the flock's mean-spread $\bar{S}$ for the same (D, N) conditions (Table IV). High values ($\bar{S} \geq 30$) tend to suggest that the dogs lost control of the flock, leading to herding failure. For a given flock size N , adding more dogs initially diminishes $\bar{S}$ and helps to increase SR. This tendency reverses when dog overcrowding occurs (e.g. $N = 10$, $D \geq 20$), leading to larger $\bar{S}$ and failure rates. Also, for larger flocks more compactness (smaller $\bar{S}$) is needed to ensure success, relative to smaller flocks (§ VI-C).

C. Correlation rankings of high-level success indicators

The mean-spread $\bar{S}$ is a relatively good predictor of task outcome (Spearman $\rho = -0.701$ across 110 conditions, indicating a strong inverse correlation between $\bar{S}$ and SR). However, $\bar{S}$ alone cannot capture the entire task difficulty. In some cases, multiple dogs can achieve low $\bar{S}$ for large flocks, yet be unable to contain or drive them (e.g. 10 dogs fail for 400 sheep, while achieving a relatively small $\bar{S} = 22$). Conversely, smaller flocks may be harder to compress yet dogs manage to herd them (e.g. 1 dog succeeds for 5 sheep with $\bar{S} = 42$).

To improve predictability, we tested diverse combinations of $\bar{S}$, D and N (Table V). Spearman correlations were computed

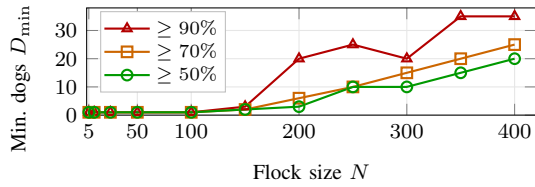


Fig. 3. Minimum number of dogs required to reach a success threshold

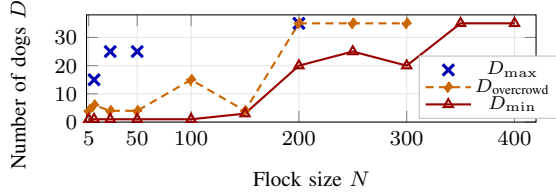


Fig. 4. Viable range $[D_{\min}, D_{\max}]$ for $SR \geq 90\%$. $D_{\min}$: smallest D reaching $SR \geq 90\%$; $D_{\text{overcrowd}}$: smallest $D > D_{\min}$ where SR starts decreasing; $D_{\max}$: smallest $D > D_{\min}$ with $SR < 90\%$ – only shown where this condition was observed (for other N , $D = 35$ ensured $SR \geq 90\%$).

using the Python library `scipy.stats.spearmanr` [23] with average rank assignment for ties (Appendix D). For each predictor, we found the optimal classification threshold by an exhaustive search over the 110 observed predictor values. We retained the cut-off that maximizes classification accuracy.

The best correlation is $\bar{S} \times N$ ($\rho = -0.828$), followed by $\bar{S} \times \sqrt{N}$ ($\rho = -0.819$). Predictors using D perform least well, probably as D is already captured indirectly by $\bar{S}$ (i.e. as more dogs imply lower $\bar{S}$). The improvement from $\bar{S}$ alone ($\rho = -0.701$) to $\bar{S} \times N$ ($\rho = -0.828$) shows that the flock size carries extra information. That is, a given $\bar{S}$ is easy to manage when the flock is within certain bounds ($N \approx [100, 200]$) yet more challenging for much larger or smaller flocks ($N \geq 300$; $N \leq 10$), as other difficulty factors intervene. The composite predictor $\bar{S} \times N$ remains a collective flock measure. In the future, we may consider additional flock metrics (e.g. shape or perimeter) to capture this dependence.

Predictor	Spearman ρ	Classification	
		< 15%	$\geq 90\%$
$\bar{S} \times N$	-0.828	107 (97%)	96 (87%)
$\bar{S} \times \sqrt{N}$	-0.819	110 (100%)	94 (85%)
$\bar{S}$	-0.701	108 (98%)	86 (78%)
$\bar{S} \times N/D$	-0.614	107 (97%)	90 (82%)

TABLE V
PREDICTOR COMPARISON (110 CONDITIONS).

D. Completion Time

Results indicate that the mean completion time on successful runs ($\bar{t}_{succ}$) increases with flock size and decreases as we add more dogs (Table VI, Figure 6). E.g., for $N=200$, $\bar{t}_{succ}$ goes from 8,100 ticks for $D=2$ to 3,400 ticks for $D=25$. We note that $\bar{t}_{succ}$ gets closer to the time limit (10,000 ticks) near

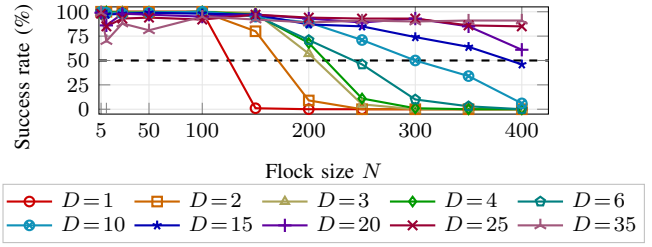


Fig. 5. Success rate based on (N, D) . 100 runs each.

breakpoints – e.g. condition $(D=1, N=150)$ only succeeded once, at 9,000 ticks. Finally, $\bar{t}_{succ}$ also increases for overly small flocks (e.g. $N \leq 10$ for $D=1$; $N \leq 25$ for $D=35$), because of less flock compression and dog overcrowding.

$N \backslash D$	1	2	3	4	6	10	15	20	25	35
5	2.0	1.6	1.4	1.4	1.4	1.6	2.1	2.5	2.7	2.7
10	1.6	1.6	1.6	1.6	1.7	2.1	2.5	3.2	3.2	4.1
25	1.6	1.9	1.8	1.8	2.0	1.9	2.6	3.1	2.8	3.5
50	2.8	2.6	2.2	2.4	2.4	2.4	2.6	2.9	2.7	3.2
100	3.1	2.4	2.9	2.8	3.2	3.2	2.9	3.4	3.2	3.4
150	9.0	5.1	3.8	3.9	4.5	3.6	3.7	3.2	3.6	3.4
200	—	8.1	6.3	5.3	5.2	4.3	3.9	3.7	3.4	3.0
250	—	—	8.9	6.2	5.8	5.2	4.5	4.0	4.2	4.0
300	—	—	—	7.8	6.5	5.6	4.9	4.5	4.5	3.8
350	—	—	—	—	8.0	6.4	6.2	5.3	4.4	4.3
400	—	—	—	—	—	7.8	6.2	5.9	4.7	4.9

TABLE VI
MEAN TICKS TO SUCCESS ($\times 10^3$, SUCCESSFUL RUNS ONLY).
WITH SUCCESS RATES $\geq 90\%$, 15–90%, < 15%.

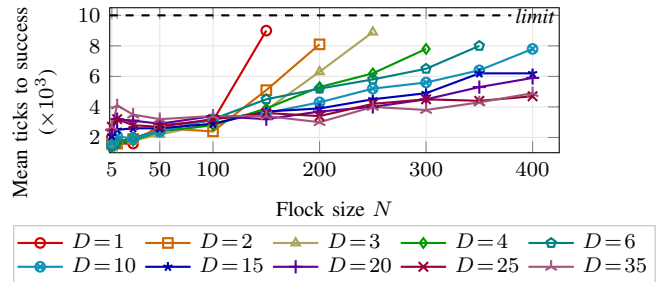


Fig. 6. Mean completion time by (N, D) on successful runs.

E. Dog Travel Distance

Table VII shows how the total distance that dogs cover ($\bar{d}_{dogs}$) rises with the dog count D . It increases and finally saturates near breakpoints, where the dogs' maximum effort becomes insufficient for collecting the flock. E.g., for the breakpoint at $N \geq 150$, $D=1$, the dog covers $\approx 10,000$ patches – its maximum over a simulation (10,000 ticks).

F. Failure analysis

Table VIII shows that 90% of overall failures occur during the collecting phase (p1) – i.e. dogs cannot compress the flock

$N \backslash D$	1	2	3	4	6	10	15	20	25	35
5	1.6	2.4	3.3	4.7	6.7	11.9	22.3	32.8	45.0	62.3
10	1.4	2.6	3.8	5.1	8.6	15.8	28.6	57.8	75.2	149.8
25	1.4	3.0	4.3	5.9	9.4	15.8	29.4	45.1	57.3	103.6
50	2.4	4.2	5.3	7.4	11.4	18.7	30.0	44.7	55.8	106.1
100	2.8	3.9	7.0	9.0	15.1	24.8	35.4	54.1	66.9	93.9
150	9.7	10.2	9.7	13.4	22.7	31.2	45.9	52.8	72.5	100.4
200	10.0	18.2	20.4	22.7	32.6	41.0	55.6	66.8	73.8	98.1
250	10.0	19.7	27.7	34.8	41.5	55.7	66.2	76.5	92.4	121.6
300	10.0	20.1	29.6	37.9	52.7	68.2	79.5	81.8	100.0	123.6
350	10.0	20.1	30.1	39.7	56.1	78.8	100.1	103.3	108.0	138.9
400	10.0	20.1	30.2	40.1	58.3	90.3	111.5	132.5	118.7	155.0

TABLE VII
MEAN TOTAL DOG DISTANCE ($\times 10^3$, ALL DOGS, 100 RUNS).
WITH SUCCESS RATES $\geq 90\%$, 15–90%, $< 15\%$.

within the containment zone, irrespective of the time taken. We subdivide failures depending on $\bar{S}$ to better assess their causes. Failures with $\bar{S} > 25$ account for 70% of failures: 97% stall in collecting, 1% in holding and 2% in exiting. We assume here that dogs are unable to compress the flock. Failures with $\bar{S} \leq 25$ are qualitatively different: only 74% occur during collecting, while the rest during holding (9%) and exiting (18%). We presume that some of these cases (10% of total failures) might succeed if provided more time ($> 10,000$ ticks).

	Fails	Collect.	Hold.	Exit.	Encl. 0
$\bar{S} > 25$ (no control)	2,161	97%	1%	2%	99%
$\bar{S} \leq 25$ (control)	928	74%	9%	18%	93%
All failures	3,089	90%	3%	7%	97%

TABLE VIII
FAILURE ANALYSIS (3,089 FAILED RUNS OUT OF 11,000).
“ENCL. 0” = RUNS THAT ENCLOSED NO SHEEP.

G. Parameter sensitivity

To estimate result sensitivity to flocking parameters (Table A1) we reduced the inter-sheep repulsion radius from $R_{\text{rep}} = 3$ to $R_{\text{rep}} = 2.5$. This directly impacts flock compactness ($\bar{S}$). We reran our experiments for four (D, N) conditions, selected within the transition or failure zone. Results (Table IX) indicate a dramatic success improvement, as lower R_{rep} leads to smaller flock span $\bar{S}$. E.g., for $(D = 6, N = 400)$ SR raises from 0% to 92%; and for $(D = 3, N = 250)$ from 5% to 100%. This strengthens our argument about using $\bar{S}$ as a macro-level success predictor, equivalent to myriad combinations of micro-configurations that impact $\bar{S}$ (e.g. sheep repulsion, attraction, vision range, max turn, various weights).

H. Density-gradient validation

To evaluate the accuracy of our density-gradient approximation, we recorded for each fleeing sheep, at every tick, the angular difference $\Delta\theta$ between the gradient direction (the densest sector) and the true GCM direction. We ran one simulation per (D, N) condition (110 runs), collecting over 80 million individual observations.

D	N	$R_{\text{rep}} = 3.0$				$R_{\text{rep}} = 2.5$			
		SR	$\bar{S}$	Ticks	Dist.	SR	$\bar{S}$	Ticks	Dist.
1	150	1%	40.0	9.0	8.2	67%	31.2	5.4	4.9
3	250	5%	33.2	8.9	24.9	100%	22.0	2.7	6.9
10	300	50%	19.0	5.6	49.8	93%	17.1	3.6	28.9
6	400	0%	56.4	—	58.3	92%	20.9	4.5	22.9

TABLE IX
EFFECT OF REDUCING THE SHEEP REPULSION RADIUS.

Table X shows the gradient accuracy for each flock of size N (pooled across the 10 different dog counts). The columns signify: Obs – the number of observations (one per sheep per tick); $|\Delta\theta|$ – the mean absolute angular error between the direction towards the GCM and the density gradient sector; Correct – how often the correct sector is selected ($|\Delta\theta| \leq 22.5^\circ$); and $\pm 1 \text{ adj}$ – how often either the correct or adjacent sector are selected ($|\Delta\theta| \leq 45^\circ$).

For $N \geq 100$, $|\Delta\theta|$ ranges between 18° and 22° . $\geq 65\%$ of the approximations are in the correct sector and $\geq 87\%$ are either in the correct sector or the adjacent one. The gradient is particularly accurate in the range $100 \leq N \leq 200$ with $\geq 73\%$ for the correct sector and $\geq 92\%$ within one sector of the true direction. We observe that $|\Delta\theta|$ increases as the size of the flock lowers ($|\Delta\theta| > 30$ for $N \leq 10$, and $|\Delta\theta| > 20$ for $N \leq 50$). When the flock is sparser the density signal is weaker. A similar pattern occurs for larger flocks ($|\Delta\theta| \geq 20.5$ for $N \geq 250$), where sheep are limited by their vision range.

N	Obs. ($\times 10^6$)	$ \Delta\theta $ ($^\circ$)	Correct (%)	$\pm 1 \text{ adj.}$ (%)
5	0.1	38.1	50.7	72.3
10	0.2	30.0	60.5	81.4
25	0.3	27.1	62.7	83.9
50	0.7	23.8	65.9	87.8
100	1.2	18.2	76.7	93.5
150	5.8	18.3	74.5	93.4
200	9.1	18.5	73.4	92.8
250	11.8	20.5	67.9	90.0
300	13.4	21.0	66.7	89.2
350	12.5	22.2	64.7	87.3
400	16.5	21.2	66.6	88.9

TABLE X
DENSITY-GRADIENT ACCURACY BY FLOCK SIZE N .

According to Figure 7, when there are enough dogs for a given flock ($N > 100$), we have $|\Delta\theta| < 0.5$ (i.e. sheep select the correct sector). Conversely, when dogs are insufficient, $|\Delta\theta| > 0.5$ (i.e. sheep may also pick the adjacent sector). For smaller flocks ($N \leq 50$), the flock is too scattered for the gradient approximation to be as accurate (i.e. sheep may pick sectors oriented away from the GCM). Figure 8 presents the full distribution of sector errors. For $N \geq 50$, over 87% of observations fall within the correct or adjacent sector. It climbs to over 90% for $100 \leq N \leq 250$. Errors of 3 or more sectors are rare and remain $\leq 1\%$ for $N \geq 100$.

VII. DISCUSSION

Obtained results provide a base for answering our research questions. They indicate how the number of dogs D scales

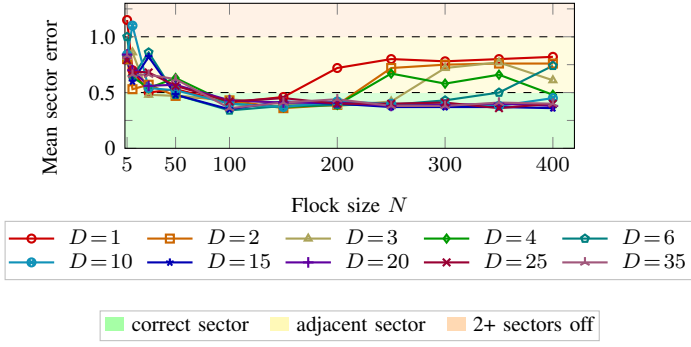


Fig. 7. Mean gradient error in number of sectors by flock size.

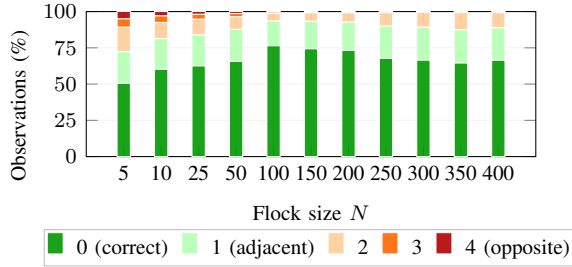


Fig. 8. Distribution of gradient error in number of sectors.

with the flock size N (**RQ1**); and correlate this tendency to high-level metrics (**RQ2**) (e.g. the flock's size and mean spread ($\bar{S}$); the dogs' effort as running distance $\bar{d}_{\text{dogs}}$ and time $\bar{t}_{\text{succ}}$). Moreover, they show how the sheep density gradient approximates the flock's Global Centre of Mass (GCM) (**RQ3**).

A. How dog numbers scale with flock size and spread

a) Non-linear scaling and diminishing returns: We note a non-linear, quasi-monotonically increasing scaling effect for the minimum number of dogs required to herd growing flocks (Figure 3). For flocks of up to 100 sheep a single dog suffices for successful herding, though small flocks of 5 to 10 sheep put more strain on the dog. For larger flocks of up to 200 sheep the dogs scale supra-linearly with the number of sheep (e.g. 3 dogs needed for 150 sheep; and 20 dogs for 200 sheep – hence 1 dog per 50 sheep, then 1 dog per 10 sheep, respectively). Finally, for flock sizes up to the maximum tested, the scaling slope strongly diminishes, appearing only slightly supra-linear (e.g. 25 dogs for 250 sheep; 20 dogs for 300 sheep; and 35 dogs for 350 and 400 sheep – hence 1 dog for ≈ 10 -15 sheep).

We also note that success rates diminish for flocks larger than 100 sheep, with $91\% \leq SR \leq 99\%$ for $N \geq 150$ sheep. This makes it difficult to compare scaling tendencies precisely in these cases. Still, we note that adding more dogs pushes the breakpoint to larger flocks, yet with a diminishing effect. This further suggests that flocks beyond a certain size may no longer be manageable and require new herding strategies (e.g. alternative dog herding, coordination or flock splitting).

b) Flock compactness helps herding: The observed scaling tendency seems to depend on the dogs' ability to limit the flock's spread ($\bar{S}$), at the expense of total distance run ($\bar{d}_{\text{dogs}}$)

and time taken ($\bar{t}_{\text{succ}}$). In the tested context, 1 dog has the easiest time herding $N=25$ sheep (compressing the flock to $\bar{S}=17.6$, while running $\bar{d}_{\text{dogs}}=1,400$ patches, for $\bar{t}_{\text{succ}}=1,600$ ticks). As the flock gets smaller, the dog has a harder time compressing it, while working harder. E.g., dividing the flock size by 5 (from $N=25$ to $N=5$) more than doubles its spread ($\bar{S}$ from 17.6 to 42), while the dog runs 14% more. Even smaller flocks may cause the dog to fail (future work).

Conversely, for larger flocks, the dog runs more and more while compressing the flock less and less. It reaches its maximum capacity at $N=100$ sheep (still achieving full success $SR=100\%$, yet with larger spread $\bar{S}=27.3$ and more effort $\bar{t}_{\text{succ}}=3,100$, $\bar{d}_{\text{dogs}}=2,800$). Hence, relative to $N=25$, x4 more sheep are only compressed half as much, while running almost twice as much. For $N \geq 150$, the dog fails abruptly ($SR=1\%$), as sheep scatter ($\bar{S} \geq 49.6$) even though the dog runs for 97% of the time. For even larger flocks ($N \geq 200$) the dog never succeeds, while running all the time – more dogs are needed.

c) Compactness is necessary but insufficient: Several cases revealed that the flock's spread alone was insufficient to forecast herding success. One case occurred when $D=10$ dogs were largely insufficient to herd $N=400$ sheep ($SR=6\%$), while they obtained a relatively compact flock, $\bar{S}=22.0$. In comparison, a single dog achieved full success ($SR=100\%$) with much larger spreads, e.g. $\bar{S}=42$ for $N=5$ sheep and $\bar{S}=27.3$ for $N=100$ sheep. In the observed case ($D=10$, $N=400$), dogs are unable to push the sheep towards the containment zone, so sheep tend to get cornered against the world boundary (where they are easier to contain). This means that the flock is compacted yet cannot be moved into the containment zone; and 88% of failed runs do not pass the collecting phase.

Other cases where compactness is insufficient for herding involve too many dogs for small flocks (cf. below). E.g., $D=35$ dogs compact $N=10$ sheep at $\bar{S}=21.4$, yet only achieve $SR=70\%$ – dogs interfere with each other and cannot guide the flock; with 93% of failures occurring in the collecting phase.

d) Dogs overcrowding: For each flock size, adding more dogs initially helps to better compress the flock. Then, adding more dogs causes overcrowding and more flock scatter. E.g. for $N=25$ sheep, going from $D=1$ to $D=10$ only reduces the spread by $\approx 14\%$ ($\bar{S}$: 17.6 $\rightarrow$ 15.2), yet multiplies the running distance by x11 ($\bar{d}_{\text{dogs}}$: 1,400 $\rightarrow$ 15,800 patches) and increases completion time by 19% ($\bar{t}_{\text{succ}}$: 1,600 $\rightarrow$ 1,900 ticks). Using even more dogs further increases the flock spread: $D=35$ dogs only manage $\bar{S}=18.2$, which is +3% more than for a single dog, while success drops by 12pp (percentage points).

e) Herding effort and breakpoints: The total dog travel distance $\bar{d}_{\text{dogs}}$ (Table VII) provides a different scaling perspective. Near breakpoints, this distance gets close to the theoretical maximum (i.e. 10,000 patches per dog), meaning that dogs run throughout the experiment (i.e. 10,000 ticks max). When dogs reach this maximum distance without managing to confine and drive the flock, the herding task necessarily fails – more dogs are needed (while considering coordination issues).

Conversely, once reliable herding is reached, adding more dogs increases their running distance without improving the

outcome. E.g., for $N=100$ sheep, a single dog achieves 100% success while only covering $\bar{d}_{dogs} = 2,800$ patches. Using 35 dogs covers 33 times more distance ($\bar{d}_{dogs}=93,900$ patches), yet lowers success to 95%. Hence, the distance metric ($\bar{d}_{dogs}$) captures two phenomena: (1) insufficient control, when dogs run at maximum capacity but fail to contain the flock; and (2) superfluous effort, when more dogs add futile running cost.

f) *Viable scale range*: Results suggest that, for the proposed flocking and herding model, there is a limited range of viable dogs-to-sheep proportions that reliably lead to herding success ($SR \geq 90\%$). Experiments performed best for flocks of $\approx 10 \leq N \leq 150$ sheep, herded by $1 \leq D \leq 25$ dogs. These ranges gave the highest success rates (Table III), low flock spreads $\bar{S}$ (Table IV), good dog efficiency (Tables VII, VI) and most accurate density-gradient approximations (Table X). Beyond this scaling area, difficulties appear in both directions. Larger flocks require more dogs, which lead to coordination issues. Smaller flocks feature weaker density gradients, which global flock cohesion relies upon. This, in turn, impacts the dogs' herding ability, which relies on flock cohesion.

g) *Generalization of the scaling principles*: Analysing parameter sensitivity shows that scaling results are tied to selected parameters – e.g. sheep repulsion (§ VI-G); also confirmed by preliminary testing of attraction and alignment (not shown here). Yet, as herding success can be linked to higher-level phenomena and metrics, the general findings reported are expected to hold across a wide range of parameter combinations. This includes the non-linear scaling of dog numbers with the flock size and its correlation with the flock's spread ($\bar{S}$) – this latter is highly influenced by flocking parameters (Table IX), making it a good aggregate indicator. Other general phenomena include the dog overcrowding effect and the difficulty of herding small flocks. We also expect the sheep density gradient to apply for approximating the flock's centre as long as the flock remains within certain spread limits. Finally, while different algorithms, environments or herding objectives may impact the actual scaling breakpoints, we believe that any setting will feature a limited scaling range.

B. How the sheep gradient approximates the flock's centre

Results in Fig. 7 indicate that the sheep density gradient works best when dogs are in sufficient numbers to keep the sheep relatively compact – i.e. for $D \geq 6$ dogs and $N \geq 200$ sheep, the error is of maximum one sector, $|\Delta\theta| \leq 45^\circ$. Even when sheep become more scattered they still manage to guess the general direction of the GCM – i.e. maximum error of two sectors, $|\Delta\theta| \leq 90^\circ$, hence never heading in the opposite direction. This tendency is reversed for small flocks ($N \leq 50$), where sheep are too few or scattered to form a gradient. Here, the direction error can sometimes reach three sectors (i.e. $90^\circ \leq |\Delta\theta| \leq 135^\circ$, for $D=1$ dog and $N=5$ sheep).

VIII. CONCLUSIONS AND FUTURE WORK

This paper aimed to study how exogenous control scales with the number of individuals in a collective movement

situation. The study focused on the specific example of sheep herding with shepherd dogs. The tested scenario required dogs to collect the sheep and drive them through an exit gate; within a time limit. Results indicated that the number of dogs required to complete the task scaled non-linearly with the number of sheep in the flock. Namely, the number of dogs required generally increased for larger flocks, yet the scaling slope differed depending on the range of flock sizes. This was probably due to multiple interacting and competing factors affecting the herding task, leading to non-linearity and causal feedback loops – e.g. flock size and cohesion influencing the dogs' efficiency, in turn influencing the flock's spread, impacting required dog efforts to limit the flock spread; also conjugated with dog coordination and overcrowding effects.

In future work we aim to extend this study to determine more precisely the causal principles behind these scaling tendencies. This includes considering larger flocks and more dogs to determine whether there is a maximum flock size beyond which no number of dogs can accomplish the herding task (i.e. scaling-up break-point, due to diminishing returns on extra dogs). Extending the time limit for herding large flocks may be necessary here – especially for cases where our experiments failed during the last herding phase.

Similarly, we will explore the kinds of tasks that dogs can achieve for flocks of less than five sheep (i.e. scaling-down break-point, due to lack of flock coherence). For viable flock sizes, we will further analyse the most suitable dog ranges and flocking configurations, ensuring most reliability with least effort. We aim to integrate more sophisticated algorithms for flocking [2], [16], escaping [5], herding [8], splitting [7] or 'shedding' [11]. Heterogeneous algorithms for dogs and sheep may also be considered; as well as their efficacy with diverse tasks and environments. To better correlate herding success to higher-level indicators we will explore alternative aggregate metrics– e.g. the flock's perimeter or shape; or using DBSCAN [24] when flocks split into clusters.

This study provides promising indicators and research directions for developing our comprehension of scalability trends in control schemes for diverse collective movement systems.

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APPENDIX

A. Model parameters table

Table A1 lists all model parameter values for our simulation experiments.

B. Gate Attraction

Gate attraction is only active during the exiting phase (P’). An intensity factor is applied depending on dog proximity:

$$\iota = \begin{cases} 1.5 & \text{if a dog is within } R_{\text{flee}} \\ 0.3 & \text{otherwise} \end{cases} \quad (4)$$

Gate attraction relies on the sum of three components to help sheep exit smoothly rather than piling-up at the gate:

- Right pull (magnitude $1.5 \times \iota$) for sheep that are near the gate (within 3 horizontal and 5 vertical patches).
- Strong pull (magnitude 5.0) for sheep that are within 5 horizontal patches and vertically-aligned with the gate.
- Vertical correction for sheep that are within 10 horizontal patches from the gate and vertically outside of the opening:

$$F_{\text{vert}} = 0.6 \iota \times \underbrace{\left(2 - \frac{d_x}{5}\right)}_{\text{Gate proximity}} \times \underbrace{(1 + 0.3 \Delta y)}_{\text{Distance from opening}}$$

where d_x is the horizontal distance to the gate and Δy is the vertical distance from the gate opening.

C. Dog Movement Adaptation Details

- *Inter-dog repulsion*: pushing dogs away from other dogs (weight $W_{\text{dog-rep}} = 0.8$ and radius $R_{\text{dog-rep}} = 5.0$).
- *Sheep avoidance*: repelling dogs from sheep closer than $R_{\text{flee}} \times 0.5 = 7$ patches, with magnitude $F = (7 - d_{\text{dog, sheep}})/d_{\text{dog, sheep}}$.
- *Wall avoidance*: repelling dogs within 3 patches of a wall (inverse-square), with strength 1.0.
- *Speed modulation*: slowing down dogs relative to their base speed ($v_{\text{dog}} = 1.0$) – via two mechanisms: (1) Dog proximity: if there is another dog at a distance $d < 5$, the speed is multiplied by $(0.5 + 0.5 \cdot d_{\text{dog}}/4)$. It is reduced to a minimum of 50% at contact. (2) Flock proximity: When the flock centroid is within $d_{\text{push}} \times 2.5 = 22.5$ patches, the speed is multiplied by $\max(0.2, d_{\text{flock}}/22.5)$. It reduces it down to a minimum of 20% at the centroid.
- *Stuck detection*: There is a stuck detection so that the dog changes direction after 25 ticks of no movement.

Parameter	Description	Value
<i>Sheep behaviour</i>		
R_{flee}	Fleeing radius from the dog	14.0
R_{vig}	Vigilance radius	25.2
$R_{\text{sheep-rep}}$	Sheep to sheep repulsion radius	3.0
Stress mult.	Fleeing repulsion reduction ($\times$)	0.5
Persistence	Direction persistence when grazing	0.5
n_{topo}	Topological neighbours	7.0
Fear threshold	Fleeing neighbours ratio for contagion	0.3
Vision range	Maximum viewing distance	30.0
v_{graze}	Sheep grazing speed	0.01
v_{flee}	Sheep fleeing speed	0.3
W_{flee}	Dog avoidance weight	3.5
$W_{\text{flee-gate}}$	Dog avoidance near gate	0.5
W_{rep}	Sheep repulsion weight	1.5
W_{wall}	Wall avoidance strength	5.0
Wall radius	Wall avoidance radius	8.0
$W_{\text{local-flee}}$	Local fleeing cohesion weight	1.5
$W_{\text{global-flee}}$	Global fleeing cohesion weight	0.75
Density sectors	Sectors for global cohesion	8
$W_{\text{local-graze}}$	Local grazing cohesion weight	0.5
$W_{\text{graze-align}}$	Grazing alignment weight	0.3
Max turn	Maximum turning rate (limit)	40° / tick
<i>Dog behaviour</i>		
v_{dog}	Dog speed	1.0
$R_{\text{dog-rep}}$	Dog to dog repulsion radius	5.0
$W_{\text{dog-rep}}$	Dog repulsion weight	0.8
Dog wall radius	Dog wall avoidance radius	3.0
$W_{\text{dog-wall}}$	Dog wall avoidance strength	1.0
$R_{\text{dog-sheep}}$	Dog-sheep avoidance radius	7.0
$W_{\text{dog-sheep}}$	Dog-sheep avoidance weight	1.0
d_{push}	Dog push distance from the flock centroid	9.0
Proximity slowdown	Slowdown zone ($d_{\text{push}} \times 2.5$)	22.5
d_{behind}	Dog positioning distance behind an outlier sheep	9.0
Curve radius	Curved approach radius	25.0
Curve strength	Curved approach strength	9.0
Patrol radius	Distance from containment zone	+5/+8
Formation spread	Angular dispersion behind flock when driving	90°
Fragmentation thr.	Lost sheep ratio before retreat	0.10
<i>Environment</i>		
World size		101 $\times$ 71
Holding duration	Flock containment time	800
Time limit	Maximum ticks	10,000
Density gradient	Global cohesion option	on
Lost threshold	Distance from centroid to be lost	$4\sqrt{N}$

TABLE A1
MODEL PARAMETERS

D. Predictor Threshold Details

The optimal predictor also depends on the outcome to be predicted. Of particular interest is predicting near-total failure of the herding task ($\text{SR} < 15\%$) or reliable herding ($\text{SR} \geq 90\%$). For identifying failure, $\bar{S} \times \sqrt{N} \leq 417$ classifies all 110 conditions correctly. This means that every condition above this value fails. On the other hand, for predicting reliable herding, $\bar{S} \times N \leq 3,090$ reaches 87% accuracy (96/110) with 91% precision and 86% recall (Table A2). The 5 false positives are flocks of small size ($N \leq 50$) where overcrowding of dogs ($D \geq 20$) impacts negatively the performance despite

low $\bar{S}$. The 9 false negatives are large flocks ($N \geq 200$) with numerous dogs ($D \geq 20$) that succeed in spite of a high $\bar{S} \times N$.

	Predicted $\geq 90\%$	Predicted $< 90\%$
Actual $\geq 90\%$	53 (TP)	9 (FN)
Actual $< 90\%$	5 (FP)	43 (TN)

TABLE A2
CONFUSION MATRIX FOR RELIABLE HERDING PREDICTION
($\bar{S} \times N \leq 3,090$).

E. Success thresholds detail

N	D_{min}	D_{max}	Max SR
5	1	not reached	100%
10	1	15	100%
25	1	25	100%
50	1	25	100%
100	1	not reached	100%
150	3	not reached	99%
200	20	35	94%
250	25	not reached	93%
300	20	not reached	93%
350	35	not reached	91%
400	35	not reached	91%

TABLE A3
RANGE OF DOG COUNTS REACHING $\text{SR} \geq 90\%$. “NOT REACHED” MEANS THE EXPERIMENT CEILING $D = 35$ WAS HIT WITHOUT SR DROPPING BELOW 90%, SO THE TRUE D_{max} LIES ABOVE 35. MAX SR IS THE HIGHEST SUCCESS RATE OBSERVED ACROSS ALL TESTED D FOR THAT FLOCK SIZE.